

AAYUSH K. SHARMA

Software Engineer | Backend & Distributed Systems

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Software engineer shipping greenfield production systems: a full stack legal contract analysis platform (React, TypeScript, FastAPI, Python, PostgreSQL) and two agentic microservices with tool calling, SQL execution, and retrieval pipelines, provisioned on Azure with Bicep infrastructure as code. Also scaled a 15,000 line platform to 180+ users at 0.1% downtime, reducing outages 73%. B.S. Applied Mathematics.

SKILLS & INTERESTS

Languages: JavaScript, TypeScript, Python, C++, C, SQL/PSQL, Bash, MATLAB | **Frameworks/Libraries:** Node.js, Express.js, ReactJS, Next.js, FastAPI

Tools/Cloud: PostgreSQL, Azure (IaC/Bicep, Foundry, Logic Apps), Digital Ocean, Git, GitHub Actions, Docker

Systems & Concepts: Distributed Systems, Microservice Architecture, REST APIs, CI/CD Pipelines

Agentic/AI-Assisted Development: Claude Code, Claude Design, Cursor, agentic test-driven development workflows

PROFESSIONAL EXPERIENCE

Forward-Deployed AI Integrations Engineer - Dallas, TX

Valiant Residential | Business Intelligence - May - August 2025 | June 2026 - Present

- Architected and delivered two agentic microservices with independent tool-calling, SQL execution, and retrieval pipelines during a Summer 2025 internship, engineering a persistent harness layer that standardized schema contracts and query guardrails across heterogeneous SQL backends.
- Now driving end-to-end delivery of a full-stack legal contract analysis platform (React/TypeScript + FastAPI), from product requirements investigation through architecture and deployment, enabling branch-office staff to upload contracts for automated language/compliance review and third-party financial, insurance, and business charter standing checks against external systems.
- Operating in a forward-deployed capacity, working directly with field stakeholders across multiple branch offices - internal employees engaged with an external-stakeholder-style workflow - to translate on-the-ground business needs into technical specifications for a greenfield production system.
- Practice agentic, test-driven development with tools like Claude Code, Claude Design, and Cursor - concentrating personal effort on rigorous system design, data modeling, and architecture while cogently abstracting and delegating implementation to agentic coding tools, validated through comprehensive test coverage.
- Designing and provisioning all cloud infrastructure via Infrastructure-as-Code (Bicep) on Microsoft Azure, establishing a reproducible, version-controlled deployment pipeline from the ground up.

PROJECTS & OUTSIDE EXPERIENCE

Software Engineering and Product Design Lead - College Station, TX

Album of the Week (tamuaotw.com) | Automation Committee - December 2023 - May 2026

- Senior developer and team lead over 15,000+ lines of linted, production code powering weekly club operations, scaling from 25 to 180+ members; owned end-to-end architecture decisions, code review standards, and design/BI documentation.
- Designed and built autonomous, fault tolerant background workers (Digital Ocean microservices) backed by a PostgreSQL persistence layer for weekly album nomination and voting, achieving 0.1% downtime over the past year through FMEA reviews and investment into end-to-end testing + staging environments.
- Built and maintained the full-stack web platform - Express.js backend, ReactJS frontend, Spotify RESTful API, and Google Identity - serving 50+ DAU; revamped CI/CD via GitHub Actions and authored a custom Dockerized testing environment, driving an estimated 20% increase in project velocity and reducing SEV-1 incidents by 4 events in Spring 2026.
- Overhauled security architecture and infrastructure: enforced JWT-authenticated endpoints via Google OAuth2 with PSQL Row-Level Security, and migrated legacy Discord bot workloads to Digital Ocean managed services, reducing outages by 73%.
- Presented a year-in-review (Spotify Wrapped-esque) exploratory data analysis to ~100 non-technical members, dissecting listening patterns and music data features around the question: "how did music impact members this semester?"

Founder & Sole Developer - Dallas, TX

"2nd Brain" - Personal Knowledge Graph App - May 2026 - Present

- Architecting a local-first, personal knowledge-graph application in TypeScript for durably tracking learning over time, modeling heterogeneous sources (video, articles, PDFs, AI chat sessions), precise in-source anchors, and Markdown notes as nodes within a unified, typed edge graph.
- Designed a SQLite persistence layer using a registry pattern - a central node registry for identity/universal fields with per-type tables for the rest - enforced behind a strict "save a node" API boundary, with user-managed labels as first-class entities across nodes and edges.
- Engineered a two-layer durability system to guard against source rot: mandatory bounded excerpts at every anchor (text in DB, media in a content-addressed blob store), plus tiered source archives governed by per-type and global storage budgets that warn and refuse rather than silently evict.
- Built as a local Node/Bun HTTP API server with a browser-based frontend and a sparse, vim-centric editor on CodeMirror 6, deferring an Electron/Tauri desktop shell to a later step to keep the dev loop fast.

EDUCATION

Texas A&M University - College Station, TX

B.S. Applied Mathematics, Computer Science Emphasis - August 2022 - May 2026

- GPA: 3.62 | Relevant Courses: (Graduate) Math Statistics | Linear Algebra II | Data Structures and Algorithms (C++)
- Coursework: Numerical Methods (iterative, non-linear, convex optimization; linear/semidefinite programming); (Grad) Optimization I & II